3. The theta group.

We model our exposition on section 2. To save space, we resort to outline form. By definition:

$$\Gamma = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL(2,\mathbb{Z}) \ : \ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv E \ \text{or} \ I \mod 2 \right\}$$

$$\overline{\Gamma} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}) \ : \ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv E \ \text{or} \ I \mod 2 \right\} .$$

Cf. Rankin[1,pp.29,219]. It is well known that $\overline{\Gamma}$ coincides with the subgroup of $SL(2,\mathbb{Z})$ generated by E and S^2. Cf. Lehner[1,pp.353,365,366] and Rankin[1,p.63]. A trivial analysis shows that:

$$\Gamma = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL(2,\mathbb{Z}) \ : \ ab \equiv cd \equiv 0 \mod 2 \right\}$$

$$\overline{\Gamma} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}) \ : \ ab \equiv cd \equiv 0 \mod 2 \right\} .$$

Review chapter 8 section 2. Take $\mathcal{F} = \{z \in H : |z| > 1 \ , \ |x| < 1\}$. This $\mathcal{F}$ is NOT canonical but it IS convenient.⊞ Cf. Gunning[1,p.17] and Lehner[1,p.365]. Γ has index 3 in $PSL(2,\mathbb{Z})$. Furthermore:

$$PSL(2,\mathbb{Z}) = \sum_{j=0}^{2} \Gamma R^j \qquad \text{where} \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = ES$$

$$\xi_1 = i \qquad \eta_1 = i\infty \qquad \eta_2 = -1 \qquad \mu(\mathcal{F}) = \pi$$

$$\Gamma_i = [E] \qquad \Gamma_{\eta_1} = [T_1] \qquad \Gamma_{\eta_2} = [T_2] \qquad \kappa = 2$$

$$T_1 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \qquad N_1 = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \qquad N_1 T_1 N_1^{-1} = S^{-1}$$

$$T_2 = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} \qquad N_2 = R \qquad N_2 T_2 N_2^{-1} = S^{-1}$$

$$R\Gamma R^{-1} = \left\{ \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in PSL(2,\mathbb{Z}) : \gamma \equiv 0 \bmod 2 \right\} = \hat{\Gamma}_0(2)$$

$$\Gamma R^{-1} = \left\{ \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in PSL(2,\mathbb{Z}) : \alpha \equiv \gamma \equiv \beta+\delta \equiv 1 \bmod 2 \right\} .$$

For the penultimate equation, we refer to Rankin[1,pp.25,30]. By applying Lehner[1,pp. 230-234], we see that Γ is generated by E and S^2 subject to the defining relation $E^2 = I$. Compare Lehner[1,p.365] and Rademacher[3,p.147]. With regard to the latter: note that $R^{-1}SR = S^{-2}E$ and $R^{-1}V_1R = E$ in $PSL(2,\mathbb{R})$.

LEMMA 3.1. Consider ordered pairs $\langle cd \rangle$ such that $c > 0$, $0 \leqq d < 2c$, $c+d \equiv 1 \bmod 2$, and $(c,d) = 1$. Let K_{cd} be ANY matrix in Γ whose bottom row is cd. Then:

$$\Gamma = [T_1] I [T_1] + \sum_{cd} [T_1] K_{cd} [T_1]$$

as a DISJOINT union of double cosets.

Proof. Elementary. ■

LEMMA 3.2. Consider ordered pairs $\langle cd \rangle$ such that $c > 0$, $0 \leqq d < c$, $c \equiv 1 \bmod 2$, and $(c,d) = 1$. Let L_{cd} be ANY matrix in ΓR^{-1} whose bottom row is cd. Then:

$$\Gamma = \sum_{cd} [T_1] L_{cd} R [T_2]$$

as a DISJOINT union of double cosets.

Proof. Elementary. ■

LEMMA 3.3. Consider ordered pairs $\langle cd \rangle$ such that $c > 0$, $0 \leq d < c$, $c \equiv 0 \mod 2$, and $(c,d) = 1$. Let M_{cd} be ANY matrix in $\hat{\Gamma}_o(2)$ whose bottom row is cd . Then:

$$\Gamma = [T_2]I[T_2] + \sum [T_2]\, R^{-1} M_{cd} R\, [T_2]$$

as a DISJOINT union of double cosets.

Proof. Elementary. ∎

We propose to compute $\Phi(s)$ for $\chi_0 \equiv 1$ and $m = 0$. In this regard,[⊞] see proposition 3.7 on page 280. Note that $N_j W_{oo} N_k^{-1}$ plays a crucial role.

One should also review page 281(3.8) .

LEMMA 3.4. Let $\phi(n)$ be the usual function. Cf. Titchmarsh[4,page 6]. Then:

$$F(\xi) = \sum_{\substack{n=1 \\ n \text{ even}}}^{\infty} \frac{\phi(n)}{n^{\xi}} = \frac{2^{-\xi}}{1-2^{-\xi}} \frac{\zeta(\xi-1)}{\zeta(\xi)}$$

$$G(\xi) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{\phi(n)}{n^{\xi}} = \frac{1-2^{1-\xi}}{1-2^{-\xi}} \frac{\zeta(\xi-1)}{\zeta(\xi)}$$

for $\mathrm{Re}(\xi) > 2$.

Proof. Straightforward. Cf. Titchmarsh[4,pp.6-7]. ∎

By combining proposition 3.7 with lemmas 3.1 thru 3.4, we discover that:

$$\Phi(s) = \sqrt{\pi}\,\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)} \begin{bmatrix} 2^{1-2s}F(2s) + 2^{-2s}G(2s) & 2^{-s}G(2s) \\ 2^{-s}G(2s) & F(2s) \end{bmatrix} .$$

In other words:

(3.1) $$\Phi(s) = \sqrt{\pi}\,\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)}\,\frac{\zeta(2s-1)}{\zeta(2s)}\,\mathcal{N}(s) \qquad \text{where} \qquad \mathcal{N}(s) = \frac{1}{2^{2s}-1}\begin{pmatrix} 1 & 2^{s}-2^{1-s} \\ 2^{s}-2^{1-s} & 1 \end{pmatrix} .$$

Compare Hejhal[2,p.476(top)] remembering that $\hat{\Gamma}_0(2) = R\,\Gamma R^{-1}$.

A computation similar to pp.508–510 shows that

(3.2) $$[\text{the parabolic contribution for } \chi \equiv 1\,]$$

$$= 2\,[\text{the parabolic contribution for } PSL(2,\mathbf{Z})\,]$$

$$+ \frac{1}{4\pi}\int_{-\infty}^{\infty} h(r)\,\mathrm{Tr}\left[\mathcal{N}'(\tfrac{1}{2}+ir)\,\mathcal{N}(\tfrac{1}{2}+ir)^{-1}\right] dr$$

$$= 2\left\{ g(0)\ln(\tfrac{\pi}{2}) - \frac{1}{2\pi}\int_{-\infty}^{\infty} h(r)\left[\frac{\Gamma'(1+ir)}{\Gamma(1+ir)} + \frac{\Gamma'(\frac{1}{2}+ir)}{\Gamma(\frac{1}{2}+ir)}\right] dr + 2\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n}\,g(2\ln n)\right\}$$

$$- 2g(0)\ln 2 - 2\sum_{n=2^{\alpha}}\frac{\Lambda(n)}{n}\,g(2\ln n) .$$

For the $\mathcal{N}$ integral, we refer to pages 537 – 538 .

Note that:

(3.3) $$\varphi(s) = \left[\sqrt{\pi}\,\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)}\,\frac{\zeta(2s-1)}{\zeta(2s)}\right]^2 \frac{1-2^{2-2s}}{1-2^{2s}} \quad .$$

Review 299(3.34), 302(3.44), 442(2.12), and 284 item 12. We immediately deduce that

(3.4) $$\frac{\varphi'(s)}{\varphi(s)} = 4\ln\pi - 2\frac{\Gamma'(s)}{\Gamma(s)} - 2\frac{\Gamma'(1-s)}{\Gamma(1-s)} - 4\frac{\zeta'(2s)}{\zeta(2s)} - 4\frac{\zeta'(2-2s)}{\zeta(2-2s)} - 2\ln 2\left[\frac{1}{1-2^{2s-2}} + \frac{1}{1-2^{-2s}}\right]$$

(3.5) $$\mathcal{M}_{\mathfrak{c}} = 0 \quad .$$

The analog of 510(2.7) is obvious. ⊞ Some additional manipulation will then yield:

(3.6) $$N[|\gamma| \leq T] = \frac{4T}{\pi}\ln\left(\frac{T\sqrt{2}}{\pi e}\right) + O(\ln T) \quad ;$$

(3.7) $$N[0 \leq r_n \leq T] = \frac{\mu(\mathfrak{F})}{4\pi}T^2 - \frac{4}{\pi}T\ln T + \frac{2T}{\pi}\left(2 + \ln\frac{\pi}{4}\right) + O\left(\frac{T}{\ln T}\right) \quad .$$

Compare 511(2.12).

<u>PROPOSITION 3.5</u>. For the theta group (*) we have $\lambda_1 \geq \frac{1}{2}\pi^2$. In particular $r_1 \geq 2.164440$.

<u>Proof</u>. Let $\mathcal{F}$ be the set $\{z \in H : 0 < x < 2 , |z| > 1 , |z-2| > 1\}$. This set is an obvious fundamental region for Γ . Cf. Lehner[1,p.365]. The identification maps for $\partial\mathcal{F}$ correspond to S^2 and S^2E . Let $A = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -3 \\ 1 & -1 \end{pmatrix}$. We'll regard A as an element of $PSL(2,\mathbb{R})$. Note that $A^2 = I$ and $A(\mathcal{F}) = \mathcal{F}$. By studying the generators of Γ , we easily determine that

(*) with $m = 0$ and $\chi = 1$

(3.8) $$A \Gamma A^{-1} = \Gamma \quad .$$

In view of equation (3.5), φ_1 is a cusp form. Cf. page 510(2.7). Let $\Psi_1 = \varphi_1(Az)$ Equation (3.8) guarantees that Ψ_1 is a cusp form for $\Gamma \backslash H$. Write:

$$\varphi_1(z) = \sum_{n \neq 0} a_n(y) e^{\pi i n x} \qquad \text{and} \qquad \Psi_1(z) = \sum_{n \neq 0} A_n(y) e^{\pi i n x} \quad ,$$

Review 70(9.5) and 282 item 2. By 70(9.4), we discover that

$$\lambda_1 = \iint_{\mathscr{D}} |\nabla \varphi_1|^2 dx\, dy \qquad \lambda_1 = \iint_{\mathscr{D}} |\nabla \Psi_1|^2 dx\, dy$$

$$\begin{aligned}
2\lambda_1 &\geqq \int_1^\infty \int_0^2 \left[|\nabla \varphi_1|^2 + |\nabla \Psi_1|^2 \right] dx\, dy \\
&\geqq \int_1^\infty \int_0^2 \left[\left|\frac{\partial \varphi_1}{\partial x}\right|^2 + \left|\frac{\partial \Psi_1}{\partial x}\right|^2 \right] dx\, dy \\
&= 2 \sum_{n \neq 0} \pi^2 n^2 \int_1^\infty \left[|a_n(y)|^2 + |A_n(y)|^2 \right] dy \\
&\geqq 2 \sum_{n \neq 0} \pi^2 \int_1^\infty \left[|a_n(y)|^2 + |A_n(y)|^2 \right] \frac{dy}{y^2} \\
&= \pi^2 \int_1^\infty \int_0^2 |\varphi_1|^2 \frac{dx\, dy}{y^2} + \pi^2 \int_1^\infty \int_0^2 |\Psi_1|^2 \frac{dx\, dy}{y^2} \\
&= \pi^2 \int_{\mathcal{C}} |\varphi_1|^2 d\mu + \pi^2 \int_{A(\mathcal{C})} |\varphi_1|^2 d\mu \qquad \text{where } \mathcal{C} = [0,2] \times [1, \infty) \\
&\geqq \pi^2 \int_{\mathscr{D}} |\varphi_1|^2 d\mu = \pi^2 \quad .
\end{aligned}$$

The desired result follows immediately. ■

This result is due to Roelcke[3,pp.50-51].

At the present time there are no experimental results corresponding to 513(2.13).

We now SHIFT our attention to automorphic forms. Let

(3.9) $$\theta(\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i \tau n^2} \quad \text{for} \quad \tau \in H .$$

It is well known that:

(a) $\theta(z)$ is an automorphic form of weight $\frac{1}{2}$ over $\Gamma \setminus H$;

(b) $\theta(z) \in \mathcal{C}(\frac{1}{2}, v)$ where $v(S^2) = 1$ and $v(E) = e^{-\frac{1}{4}\pi i}$;

(c) $\theta(z) \neq 0$;

(d) $\theta(z) = \prod_{n=1}^{\infty} (1-q^{2n}) \prod_{n=1}^{\infty} (1+q^{2n-1})^2$ where $q = e^{\pi i z}$.

Cf. Hille[1,pp.156-157] and Rankin[1,pp.215-219]. To make (b) more precise, we review chapter 9 proposition 2.1 and observe that

(3.10) $$\theta(z) j_{N_k}(z) = \sum_{n=0}^{\infty} c_{nk} e^{2\pi i (n+\alpha_k) w} \quad \text{with} \quad c_{0k} \neq 0$$

for $w = N_k z$ and $\alpha_k = \frac{\delta_{k2}}{8}$. Cf. Rankin[1,pp.218-219] and page 505 note 30. It is understood that $j_L(z) = (cz+d)^{1/2}$. We also note that

(3.11) $$v(T_k^{-1}) = e^{2\pi i \alpha_k} .$$

Cf. 339(3.2), 492(4.17), and 426(64).

For real λ , we define $\theta^\lambda(z)$ in the obvious way and write

(3.12) $$w_\lambda(L) = \frac{\theta^\lambda(Lz)}{\theta^\lambda(z)(cz+d)^{\lambda/2}} \quad \text{for} \quad L \in \overline{\Gamma} .$$

This ratio defines a multiplier system of weight $\frac{\lambda}{2}$. Cf. Rankin[1,pp.220-221]. By substituting appropriate values for z , we see that:

(3.13) $$W_\lambda(S^2) = 1 \quad \text{and} \quad W_\lambda(E) = e^{-\frac{1}{4}\lambda\pi i} .$$

In particular:

(3.14) $$W_1 = V \quad \text{and} \quad W_\lambda \equiv 1 \quad \text{iff} \quad \lambda \equiv 0 \mod 8$$

(3.15) $$W_{\lambda_1} W_{\lambda_2} = W_{\lambda_1+\lambda_2}$$

(3.16) $$W_\lambda(T_k^{-1}) = e^{2\pi i\alpha_k(\lambda)} \quad \text{where} \quad \alpha_k(\lambda) = \left\{ \begin{array}{ll} 0 & \text{for } k=1 \\ \hline \frac{\lambda}{8} - [\![\frac{\lambda}{8}]\!] & \text{for } k=2 \end{array} \right\} .$$

We can now assert that:

(A) $\mathcal{A}(m, W_{2m}) = \mathcal{B}(m, W_{2m}) = \mathcal{C}(m, W_{2m}) = \{0\}$ for $m < 0$;

(B) $\mathcal{A}(m, W_{2m}) = \{0\}$ and $\mathcal{C}(m, W_{2m}) = \mathbb{C}\theta^{2m}$ for $0 \leqq m < 4$;

(C) $\dim \mathcal{C}(m, W_{2m}) = 1 + [\![\frac{m}{4}]\!]$ for $m \geqq 4$;

(D) $\dim \mathcal{A}(m, W_{2m}) = \lim\limits_{\epsilon \to 0^+} [\![\frac{m-\epsilon}{4}]\!]$ for $m \geqq 4$;

(E) $\mathcal{C}(m, W_{2m}) = \mathbb{C}\theta^{2m} + \mathcal{A}(m, W_{2m})$ for $m \geqq 0$ with $W_{2m} \not\equiv 1$;

(F) $\mathcal{B}(m, W_{2m}) = \mathbb{C}\theta^{2m}$ for $0 \leqq m < 1$;

(G) $\mathcal{B}(m, W_{2m}) = \mathcal{A}(m, W_{2m})$ for $m \geqq 1$.

This result agrees with Rankin[1,p.222]. The proof is not difficult. We begin by reviewing page 520 paragraph 4. Note that:

(3.17) $$\theta^{2m}(z)\,|_{N_k}(z;m) = e^{2\pi i(2m\alpha_k)w} \left[\sum_{n=0}^{\infty} c_{nkm} e^{2\pi i n w} \right] \quad \text{with} \quad c_{0km} \neq 0 .$$

In addition:

$$(3.18) \qquad 2m\,\kappa_k = \kappa_k(2m) + \delta_{k2}\,[\![\tfrac{m}{4}]\!] \quad .$$

Assertion (A) corresponds to page 505(note 31) and proposition 5.14(c). To establish (B), we consider f/θ^{2m} . Assertions (F) and (G) follow from (B) and page 505(notes 30+31) . To prove (E), we apply (3.17) and think about $f - b\,\theta^{2m}$. Cf. (3.18) and 505(note 30). Assertion (D) follows from corollary 4.2.(*) Assertion (C) follows from (E) and theorem 4.9. With regard to the latter: see 527(3.1) and 483(4.2) .$^{\boxplus}$

By considering the case $m = \frac{1}{2}$ ($K_0 = 1$) we immediately determine that E(z;s;v) has a simple pole at $s = \frac{3}{4}$. Cf. assertion (F) , proposition 5.14(d)$^{\boxplus}$, page 371(5.31) , and 374 item 17. Compare Roelcke[2,pp.304-305].

4. The congruence subgroup $\Gamma_0(N)$. $^{\boxplus}$

Let N be a positive integer. Define:

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}) : c \equiv 0 \bmod N \right\} \quad \text{and} \quad \hat{\Gamma}_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in PSL(2,\mathbb{Z}) : c \equiv 0 \bmod N \right\} .$$

Cf. Rankin[1,p.25], Schoeneberg[1,p.78], and Shimura[1,p.24]. For simplicity we assume that:

$$(4.1)^{\boxplus} \qquad N \text{ is square free} \quad .$$

It is well known that:

$$[\,\Gamma_0(1) : \Gamma_0(N)\,] = N \prod_{p \mid N} \left(1 + \frac{1}{p}\right)$$

$$n_2 = \prod_{p \mid N} \left(1 + \left(\frac{-4}{p}\right)\right)$$

(*) in chapter 10